

Project Understanding

Payment Orchestrator is a Spring Boot application that routes payment requests to the right provider, persists their state, and protects against duplicate submissions.

What it does

- Accepts create-payment and fetch-payment requests through REST APIs.
- Routes CARD traffic to PROVIDER_A and UPI traffic to PROVIDER_B.
- Uses retries, failover, and circuit breaker protection to improve resilience.

How it stores data

- MySQL keeps durable records for payments and idempotency keys.
- Redis supports fast idempotency checks and cached duplicate responses.

Why it matters

- Idempotency helps prevent duplicate charges when a client repeats a request.
- Metrics and OpenAPI make the service easier to operate, inspect, and test.

In short: the service is designed to safely orchestrate payments, recover from provider failures, and keep duplicate requests from creating inconsistent state.